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Chapter 1: PID feedback design for adaptive cruise control

1.1 Problem formulation

The system consists of a leader vehicle A and a follower vehicle B moving on the same straight lane. The objective is to control the traction/braking force of vehicle B so that the measured distance from the leader converges to a desired constant value. The leader acceleration is treated as an unknown constant during each manoeuvre, while the follower is acted on by the control force and a resistive disturbance.

$$d(t) = x_A(t) - x_B(t), \quad e_d(t) = d(t) - d_{ref}$$

Using the notation above, a positive value of e_d means that the distance is larger than the requested gap. The follower acceleration is described by the force balance

$$a_B(t) = u(t)/m_B - F_d/m_B$$

Therefore, for a constant leader acceleration and approximately constant disturbance during the considered time interval, the distance-error dynamics can be written as

$$\ddot{e}_d(t) = a_A(t) - u(t)/m_B + F_d/m_B$$

A PID action is then selected for the follower force. After differentiating once, the constant terms disappear from the equation and the characteristic polynomial of the error dynamics becomes

$$m_B s^3 + k_d s^2 + k_p s + k_i$$

The required settling behaviour is chosen with the dominant time constant tau = 0.5 s and without oscillatory poles. A simple pole assignment that satisfies this specification is

$$p^*(s) = m_B (s + 2)(s + 3)(s + 4)$$

Matching the coefficients gives the PID gains

$$k_d = 9m_B, \quad k_p = 26m_B, \quad k_i = 24m_B$$

For the numerical simulations the mass is m_B = 1500 kg, the desired distance is d_ref = 5 m, and both vehicles start from rest with the initial distance equal to the desired distance. The resulting error under a constant leader acceleration is shown in Figure 1.1. The error rises only during the transient and then returns to zero.

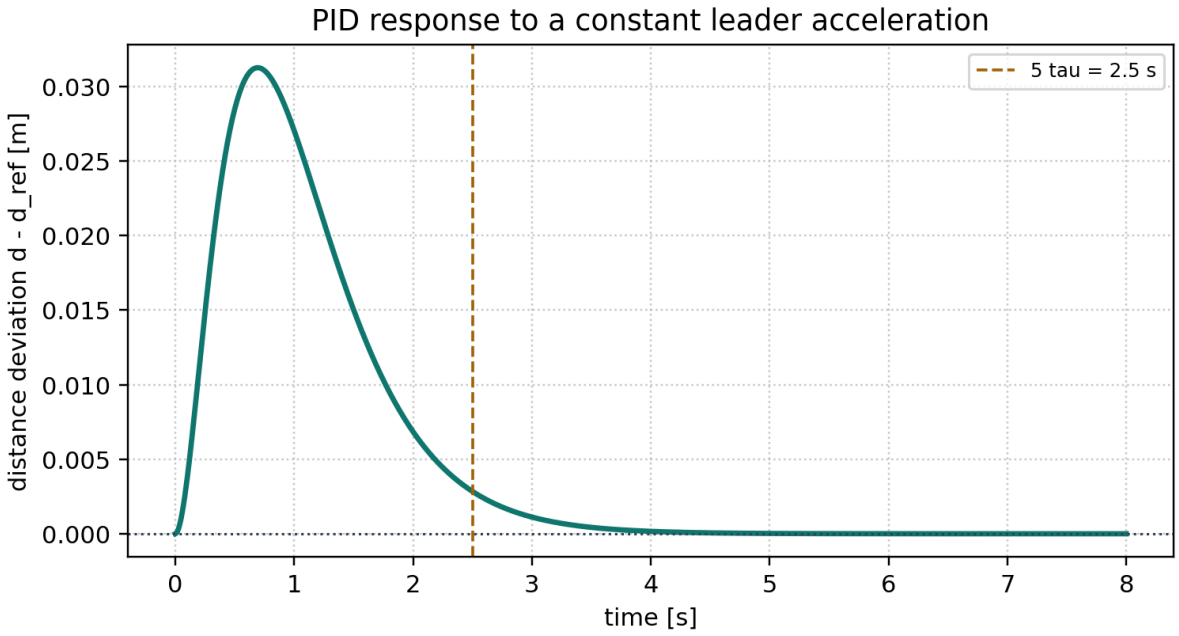


Figure 1.1: PID distance-error response for a constant leader acceleration.

Chapter 2: Output-feedback observer-based controller

2.1 Extended error model

The PID controller requires derivative information of the measured distance. In practice, differentiating the sensor signal is not robust because measurement noise is amplified. For this reason, an observer-based output-feedback structure is introduced. The state variables are selected as the distance error, the relative velocity error, and the steady force term needed to compensate the unknown constant acceleration/disturbance.

$$x_e = [\ e_d,\ e_v,\ u^*\ ]^T, \quad e_v = d_dot$$

The extended dynamics are

$$x_e_dot = A_e\ x_e + B_e\ u, \quad y = C_e\ x_e$$

$$A_e = [[0, 1, 0], [0, 0, 1/m_B], [0, 0, 0]], \quad B_e = [0, -1/m_B, 0]^T, \quad C_e = [1, 0, 0]$$

The observability matrix contains the rows C_e, C_e A_e, and C_e A_e^2. Its determinant is 1/m_B, so the system is observable for any nonzero vehicle mass.

$$\det(O) = 1/m_B \mathrel{!}= 0$$

The observer poles are placed at -10, -10 and -10, corresponding to a faster estimation time constant than the closed-loop distance-control time constant. Ackermann pole assignment gives the observer correction gains

$$L = [30, 300, 1000m_B]^T$$

The control law is a state feedback acting on the estimated states, including the estimated feedforward force:

$$u = [6m_B, 5m_B, 1]\ x_hat$$

This controller uses the estimated relative velocity rather than direct differentiation of the measured distance. Figure 2.1 summarizes the implemented structure, and Figure 2.2 shows the response under the same constant-acceleration test used for the PID design.

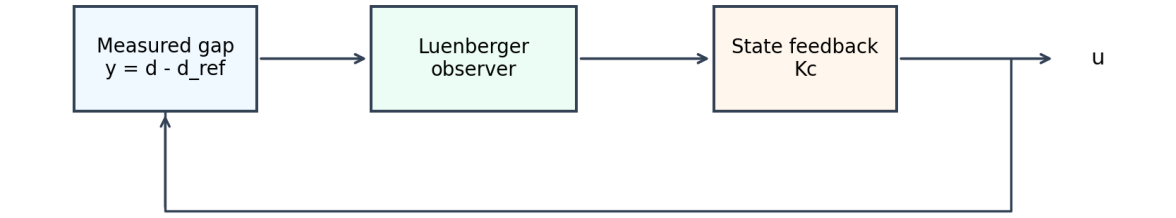


Figure 2.1: Structure of the output-feedback observer-based controller.

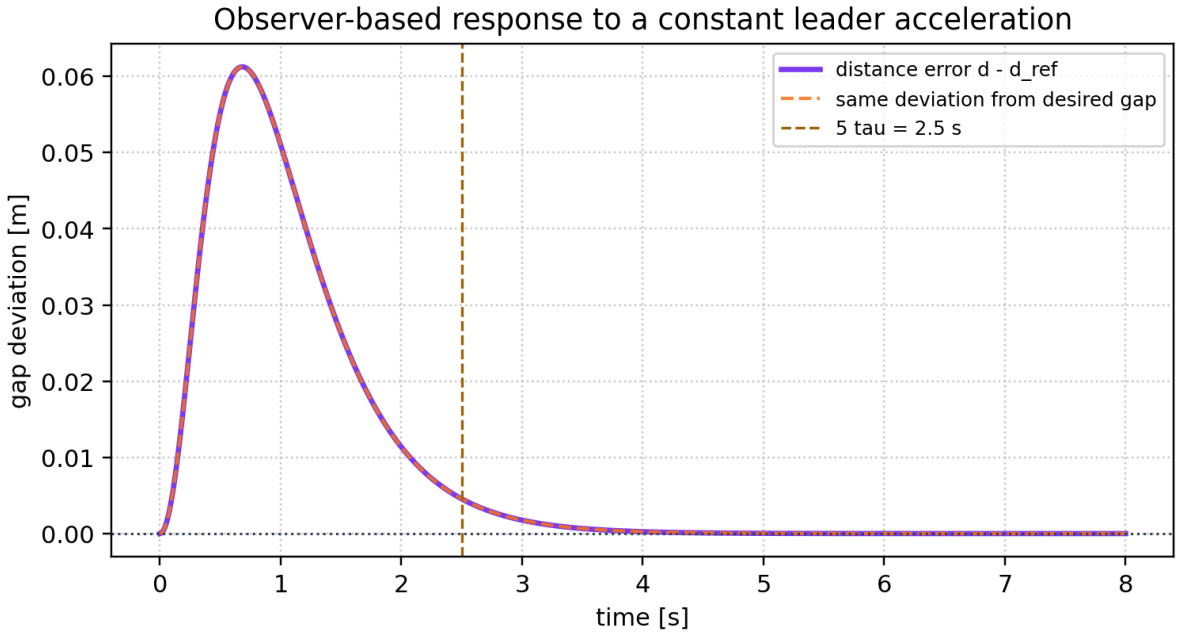


Figure 2.2: Observer-based distance-error response for constant leader acceleration.

Chapter 3: Comparison of the two control architectures

3.1 Time-domain comparison

The controllers are compared using a leader-vehicle speed profile composed of three intervals: acceleration at 1 m/s^2 for 5 s, constant speed for 5 s, and braking at -1 m/s^2 for 5 s. After that the leader remains at rest. This profile is shown in Figure 3.1.

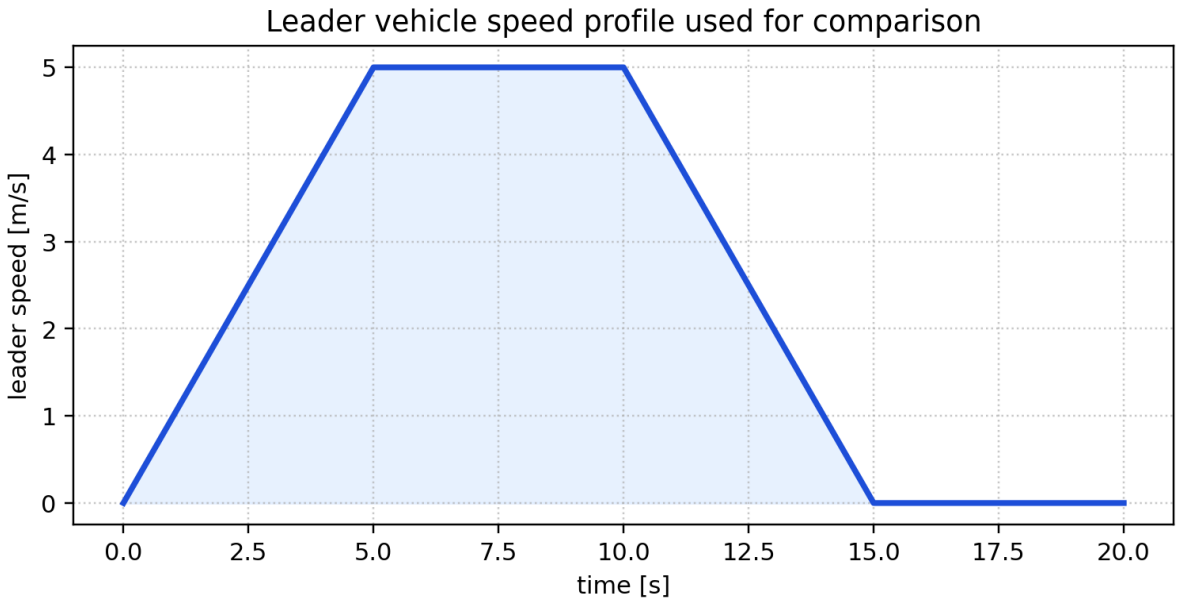


Figure 3.1: Speed profile imposed on the leader vehicle.

The PID implementation is shown in Figure 3.2. The corresponding distance-error curve is plotted in Figure 3.3. Each acceleration change creates a transient in the distance error, after which the integral action brings the steady-state error back to zero.

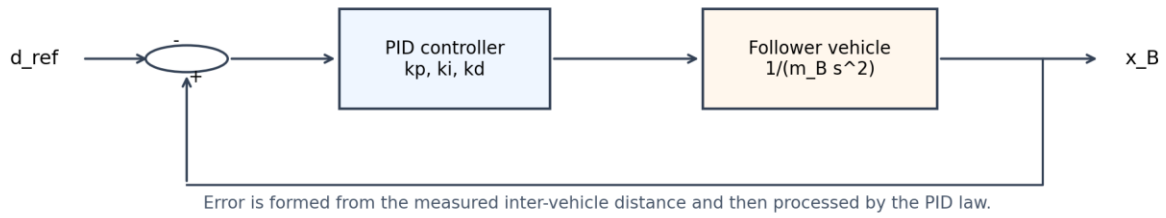


Figure 3.2: Block diagram of the PID-based distance controller.

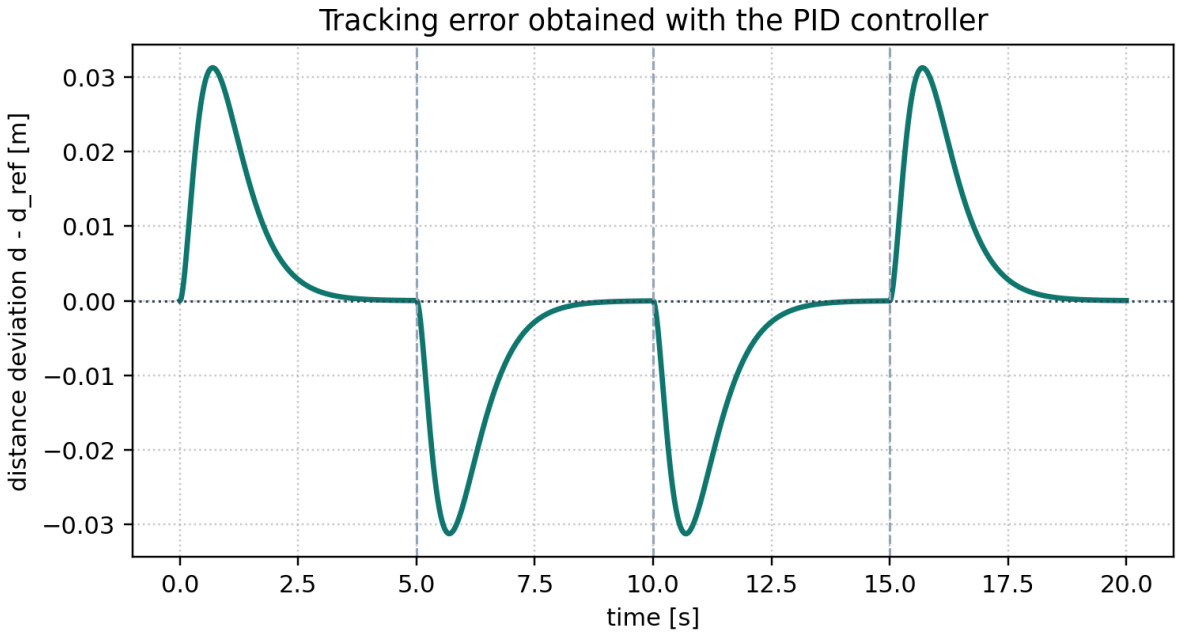


Figure 3.3: Distance error obtained with the PID controller.

The observer-based version is shown in Figure 3.4. In this case the estimated variables are used in the feedback law, and the estimated error is also plotted in Figure 3.5. The observer design produces slightly larger transient deviations, but avoids numerical differentiation of the measured distance.

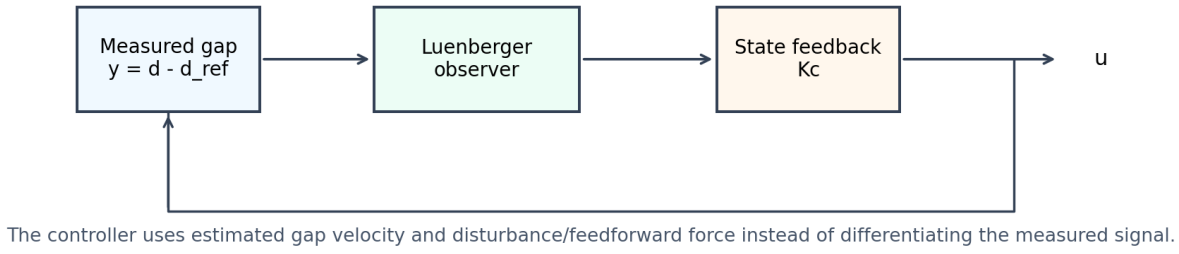


Figure 3.4: Block diagram of the observer-based controller.

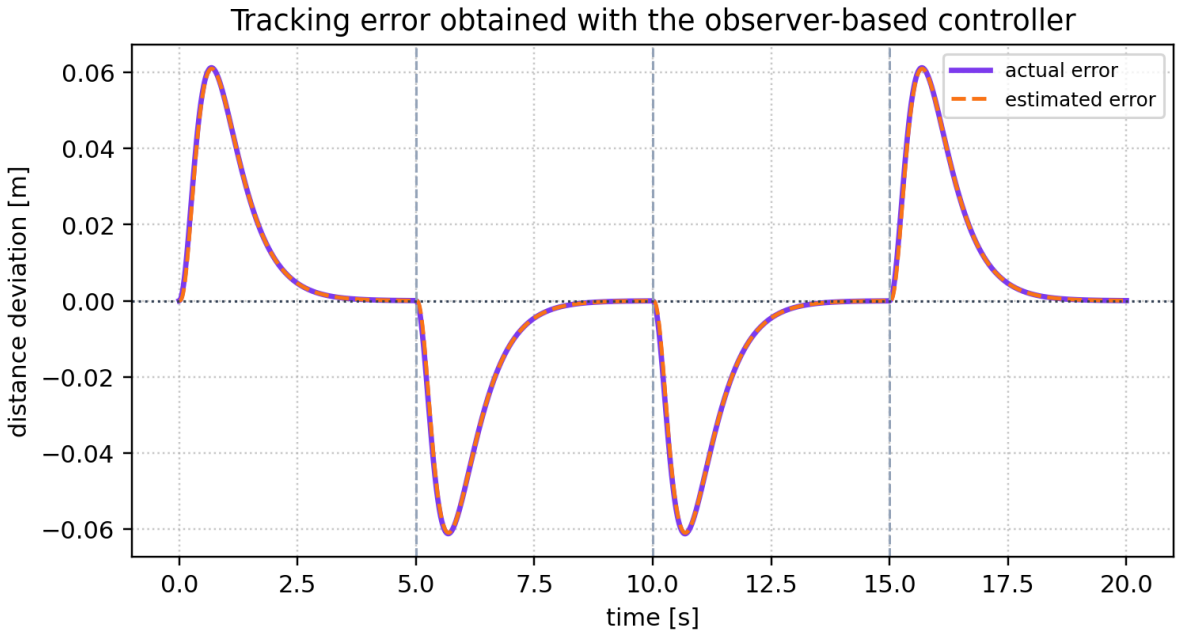


Figure 3.5: Distance error obtained with the observer-based controller.

3.2 Gang-of-four transfer functions for the PID design

The follower plant from force input to position is

$$P(s) = 1/(m_B\ s\text{^}2)$$

For the PID controller, after substituting the designed gains, the controller transfer function is

$$C_PID(s) = m_B(9s\text{^}2 + 26s + 24)/s$$

Thus, the sensitivity and complementary sensitivity functions are

$$S_PID(s) = s\text{^}3 \ / \ (s\text{^}3 + 9s\text{^}2 + 26s + 24)$$

$$T_PID(s) = (9s\text{^}2 + 26s + 24) \ / \ (s\text{^}3 + 9s\text{^}2 + 26s + 24)$$

The maximum magnitude of S_PID is approximately 1, giving a stability margin Sm = 1.0000. The resulting Bode diagrams are reported below.

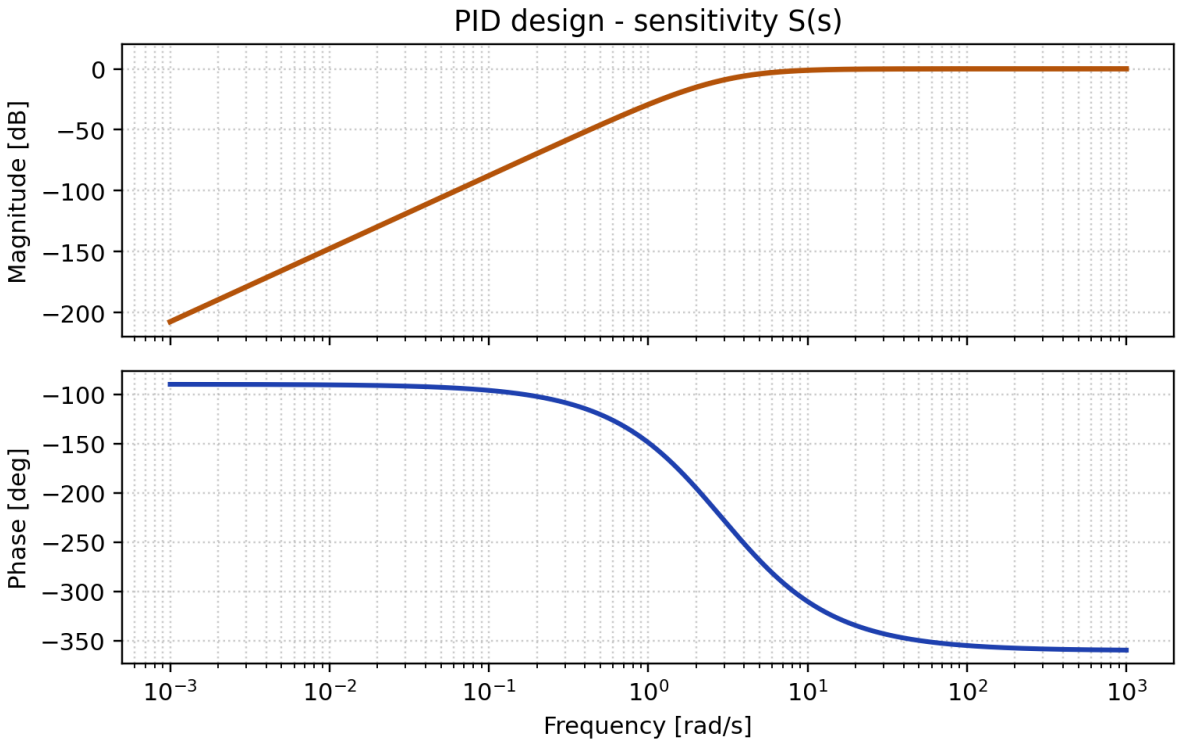


Figure 3.6: Sensitivity Bode diagram for the PID controller.

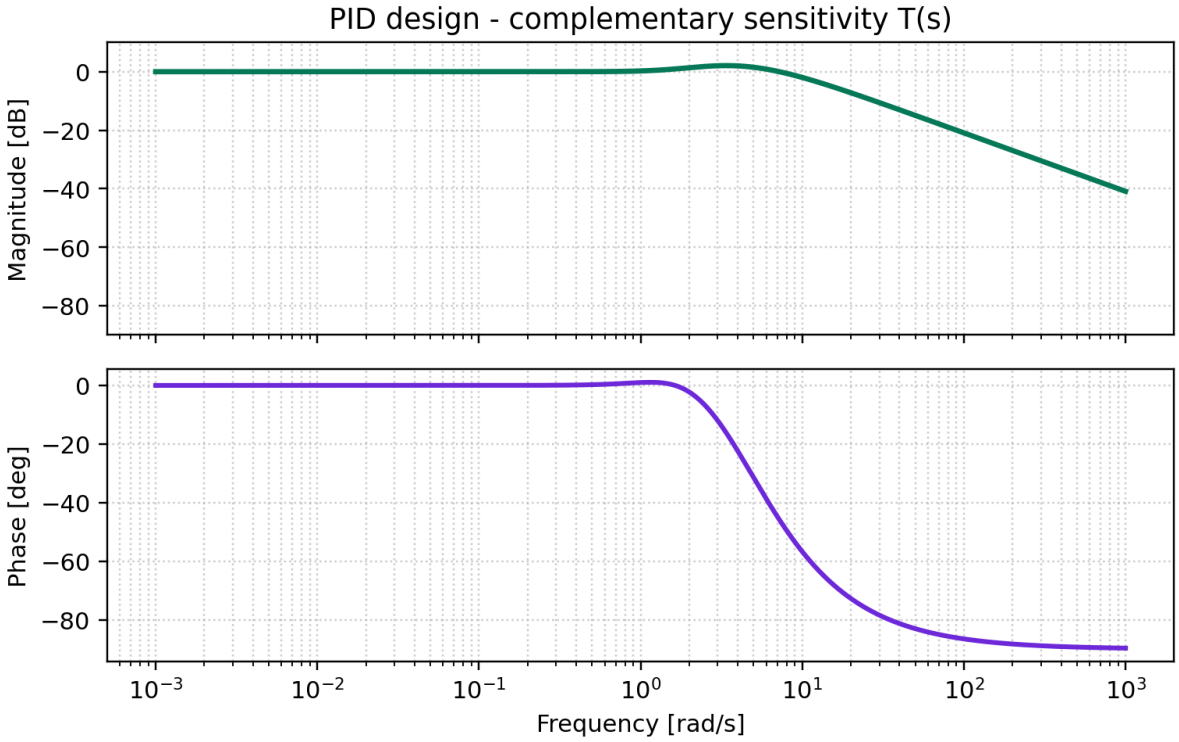


Figure 3.7: Complementary sensitivity Bode diagram for the PID controller.

The two remaining transfer functions are the load sensitivity P(s)S(s) and the control sensitivity C(s)S(s). The load sensitivity remains very small at low frequencies, which means that constant load disturbances are rejected. However, the control sensitivity of the ideal PID grows at high frequency because the derivative action is not filtered.

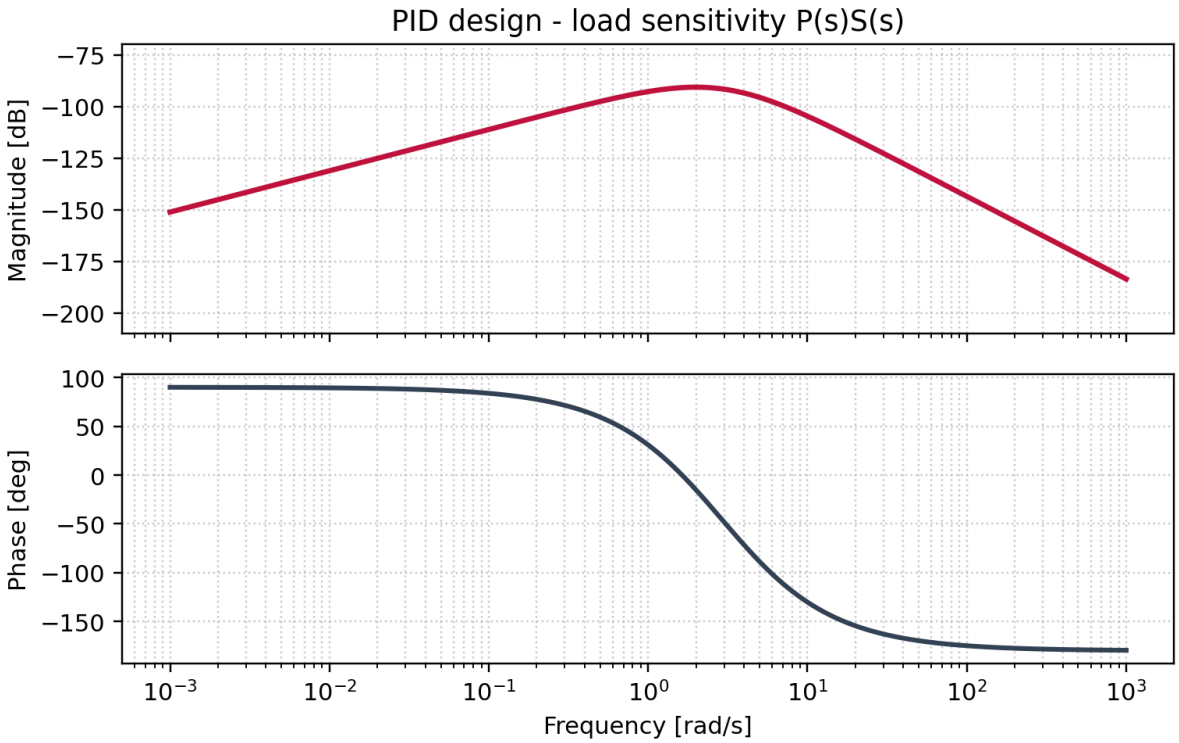


Figure 3.8: Load sensitivity Bode diagram for the PID controller.

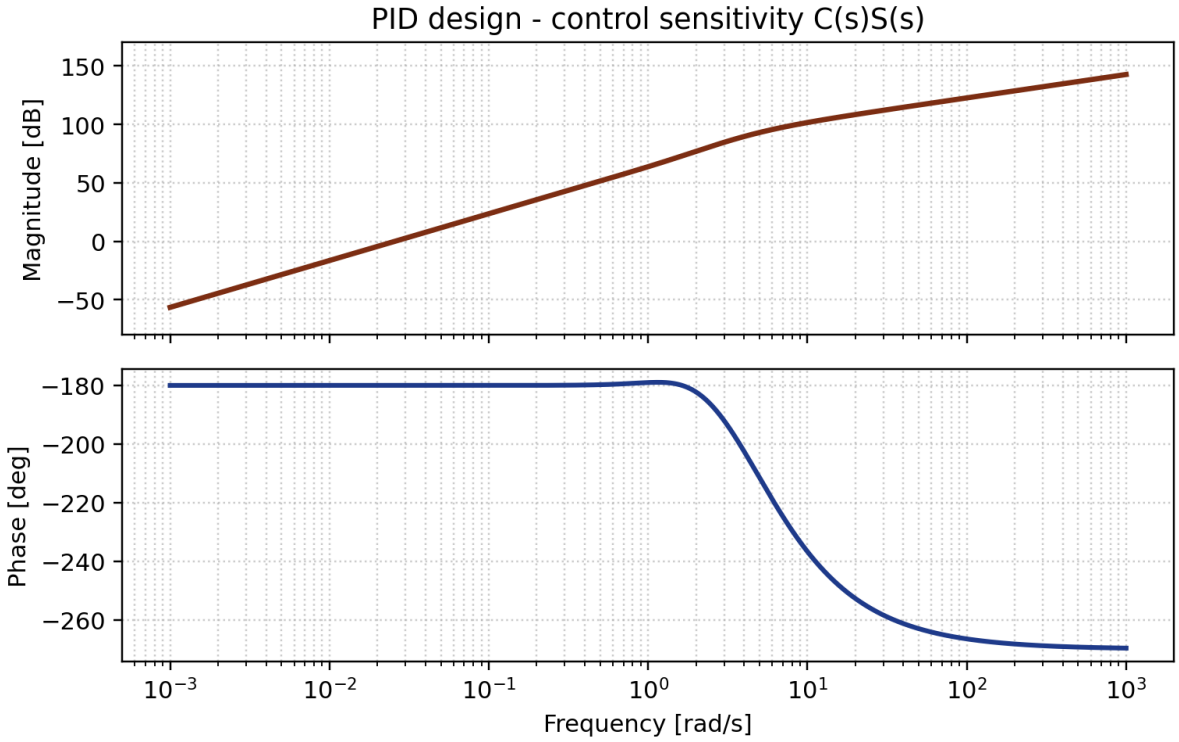


Figure 3.9: Control sensitivity Bode diagram for the PID controller.

3.3 Gang-of-four transfer functions for the observer-based design

For the observer-based dynamic compensator, the transfer function from the measured output to the control input can be written as

$$C_obs(s) = 40m_B(67s^2 + 170s + 150) / [s(s^2 + 35s + 456)]$$

Combining this dynamic controller with the same plant gives

$$S_obs(s) = s^3(s^2 + 35s + 456) / (s^5 + 35s^4 + 456s^3 + 2680s^2 + 6800s + 6000)$$

$$T_obs(s) = (2680s^2 + 6800s + 6000) / (s^5 + 35s^4 + 456s^3 + 2680s^2 + 6800s + 6000)$$

The numerical stability margin is Sm = 0.5963. It is smaller than the PID value, but the observer-based controller has the practical advantage of filtering the measured distance before generating velocity and disturbance estimates.

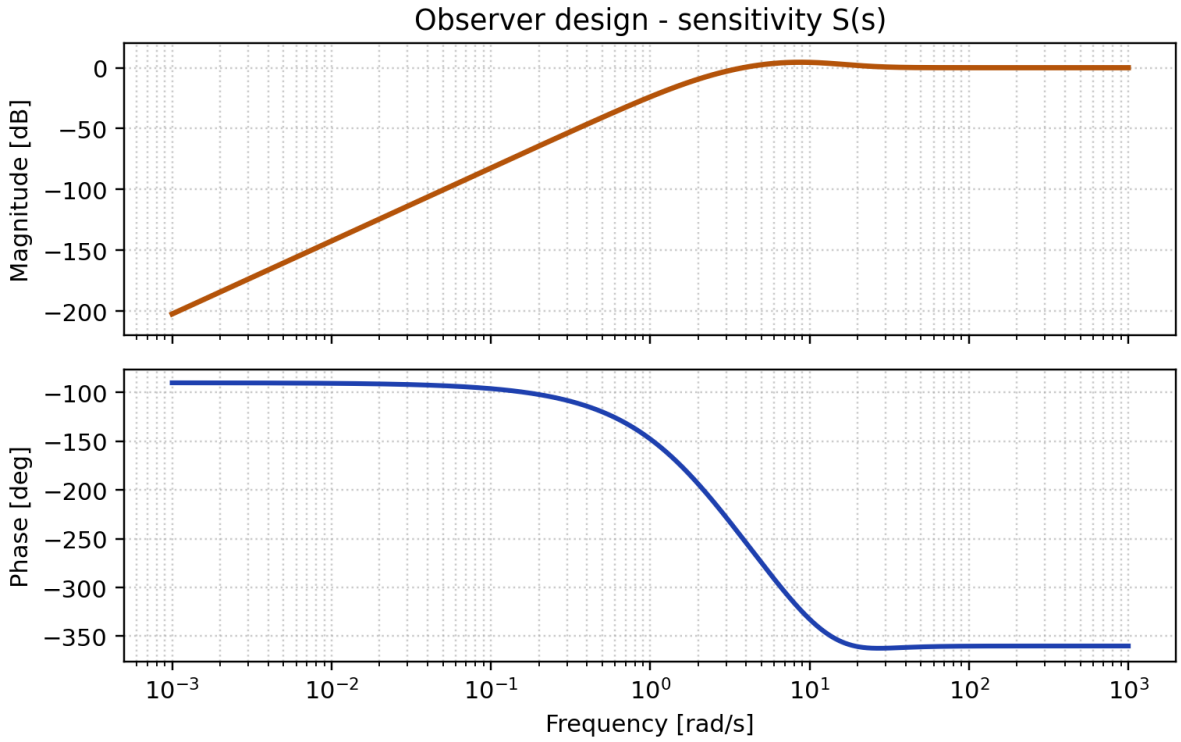


Figure 3.10: Sensitivity Bode diagram for the observer-based controller.

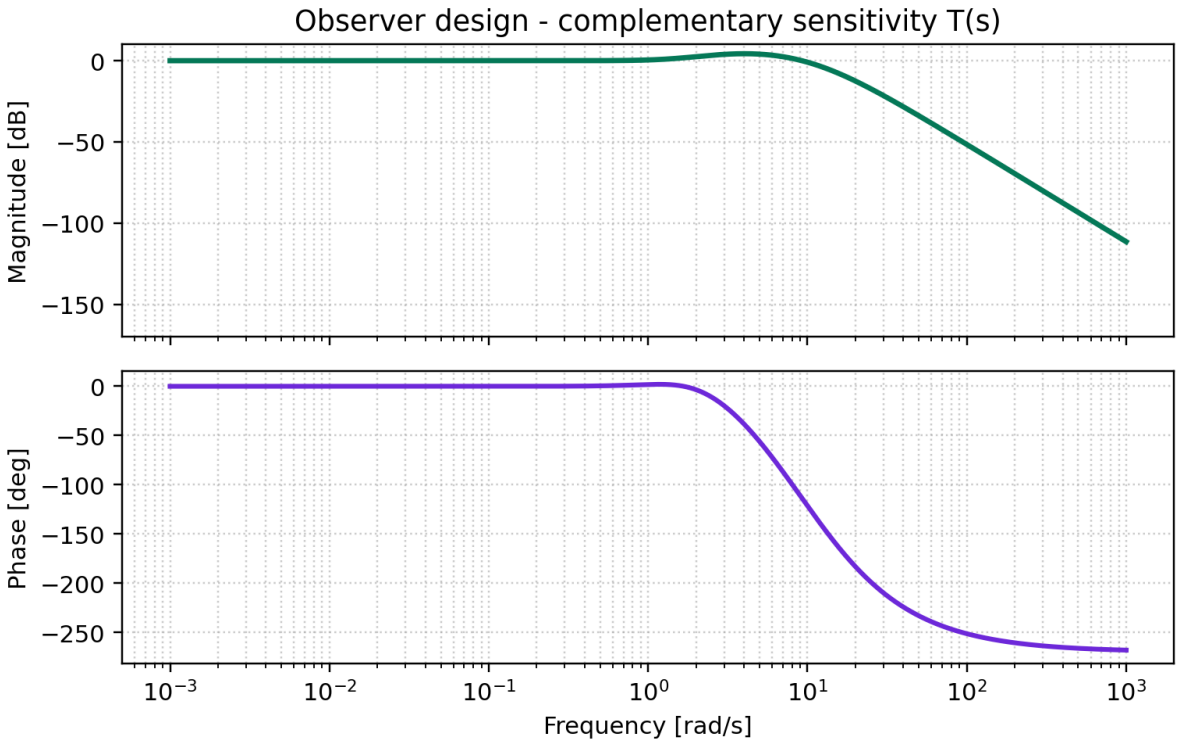


Figure 3.11: Complementary sensitivity Bode diagram for the observer-based controller.

The load sensitivity and control sensitivity are shown in Figures 3.12 and 3.13. The dynamic compensator makes the control transfer proper, unlike the ideal derivative term of the PID controller. This is one of the reasons why an observer-based structure is more realistic for noisy measurements.

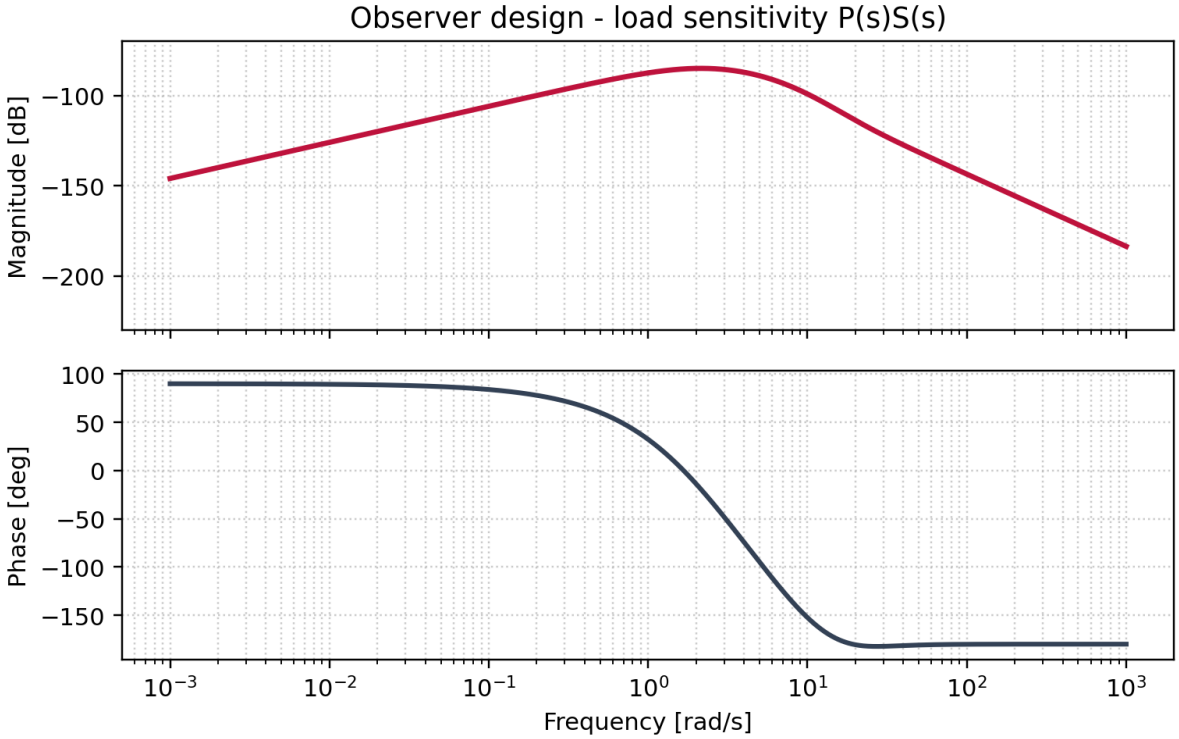


Figure 3.12: Load sensitivity Bode diagram for the observer-based controller.

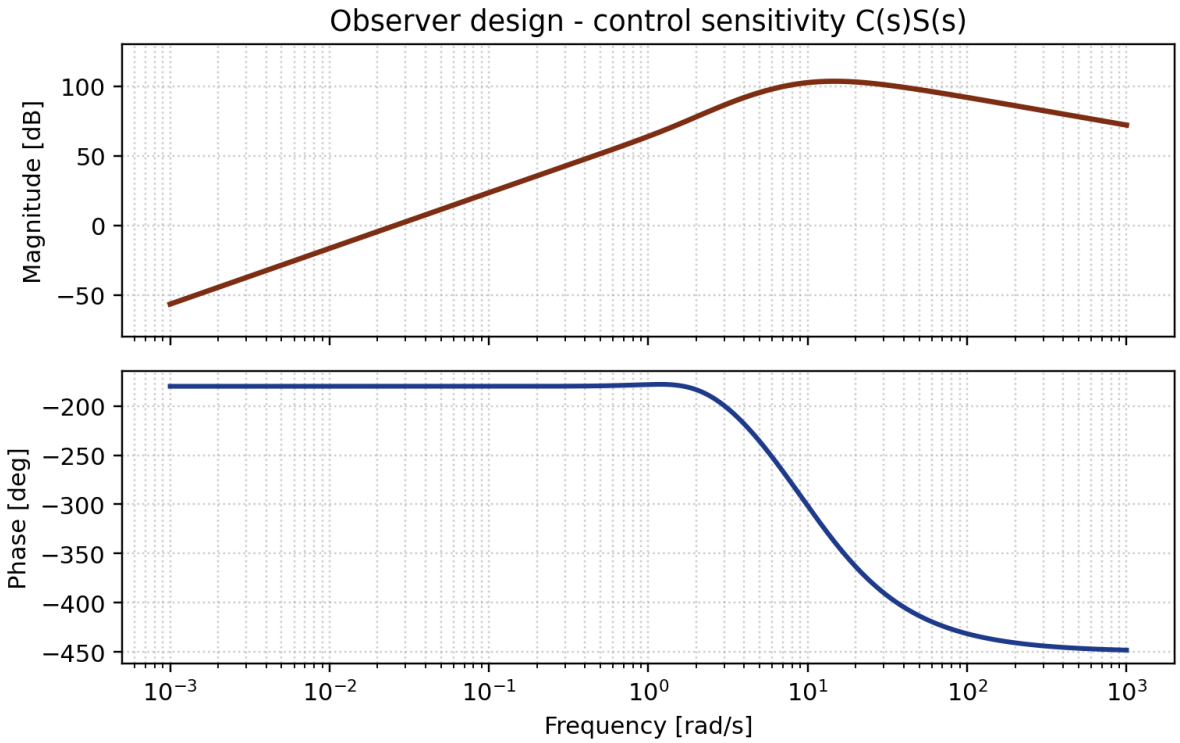


Figure 3.13: Control sensitivity Bode diagram for the observer-based controller.

Robustness discussion and conclusion

From the transfer-function analysis, the PID controller has a larger mathematical stability margin. In the ideal model this means that the PID loop is less sensitive to multiplicative uncertainty. The drawback is that the derivative action requires reliable velocity or differentiated distance information. If the measured distance is noisy, the derivative channel can amplify high-frequency noise and create an unrealistic control signal unless a derivative filter is added.

The observer-based controller gives a lower stability margin, but it is more suitable for implementation because it reconstructs the unmeasured relative velocity and the constant force/disturbance term from the measured gap. The observer behaves as a filtering element and avoids direct differentiation. Its main limitation is model dependence: if the mass or plant dynamics are inaccurate, the estimation quality can degrade. For a more realistic estimator, a Kalman filter would be preferable because it explicitly accounts for measurement and process noise in a finite-time estimation problem such as adaptive cruise control.